Loan Approval Risk Model: Project Documentation

FinTech Innovations, Risk Analytics team. Companion to the notebook `financial_loan_risk_completed_v2.ipynb`.

1. Purpose and scope

FinTech Innovations partners with banks to modernise loan approval. Manual review by loan officers is slow and inconsistent. This project builds a machine learning pipeline that scores each application with the probability that it is creditworthy, and turns that score into an approve or deny decision tuned to the company's economics:

- Approving a loan that should have been denied (false positive) loses about **$50,000** (principal plus collection costs).
- Denying a creditworthy applicant (false negative) loses about **$8,000** of profit.

The work follows CRISP-DM and the five parts of the lab brief: business understanding, data understanding, pipeline development, iterative modeling, and evaluation. The notebook headings use the lab's numbered instructions (1 to 10), so each requirement can be found directly.

Headline result. A regularised Logistic Regression pipeline reaches a test ROC-AUC of 0.996. With a cost-optimised threshold of 0.81 it costs about **$464 per applicant** in wrong decisions, against **$1,912** for denying everyone (a 76% reduction).

2. Data

Source. `financial_loan_data.csv`: 20,000 past applications, 35 columns, no duplicate rows. The target `LoanApproved` is 1 for 4,780 applicants (23.9%) and 0 for 15,220 (76.1%), an imbalance of about 3.2 to 1.

2.1 Feature groups

Group	Count	Examples	Treatment
Skewed money and ratio variables	11	MonthlyIncome, LoanAmount, NetWorth, TotalDebtToIncomeRatio	impute, standardise (log tested, rejected)
Other numeric	15	Age, CreditScore, Experience, InterestRate	impute, standardise
Binary numeric	1	PreviousLoanDefaults	pass through
Nominal categorical	4	MaritalStatus, HomeOwnershipStatus, EmploymentStatus, LoanPurpose	impute as "Missing", one-hot
Binary categorical	1	BankruptcyHistory	one binary column
Ordinal categorical	1	EducationLevel	ordinal code plus a missing flag
Target	1	LoanApproved	
Excluded (leakage)	1	RiskScore	never used as a feature

2.2 Data quality issues found, and what was done

Issue	Evidence	Action
RiskScore leaks the label	The generator script scales it by 0.8 for approved loans. A model seeing only RiskScore reaches AUC 0.987. Adding it to the other numeric features lifts AUC from 0.986 to 0.9999.	Excluded from all features. Used only as a regression target in the approach comparison.
AnnualIncome stored as text	Values such as "$39,948.00"	Cleaned to numeric on a copy of the data.
Artificial income floor and cap	584 rows (2.9%) at exactly $15,000, none approved. 26 rows at $300,000, all approved.	Flag created (IncomeAtFloor). Recommend fixing at capture.

Issue	Evidence	Action
Corrupted AnnualIncome	195 rows (0.97%) are 1.5 to 2.0 times 12 x MonthlyIncome.	MonthlyIncome treated as the reliable measure. AnnualIncome dropped after building IncomeMismatchFlag.
Informative missingness	EducationLevel is missing for 901 applicants (4.5%), and none was approved (Fisher p about 3e-110).	Ordinal code plus an explicit "was missing" flag.
Random missingness	MaritalStatus 6.7% missing, SavingsAccountBalance 2.9% missing. Neither differs significantly in approval rate.	"Missing" category and median imputation respectively.
Collinearity	BaseInterestRate is an exact combination of CreditScore, LoanAmount and LoanDuration (infinite VIF). MonthlyIncome and AnnualIncome, Age and Experience, NetWorth and TotalAssets are also strongly linked.	L2 regularisation and L1 feature selection inside the pipeline.

2.3 Strongest relationships with approval

By effect size (Mann-Whitney U, Bonferroni corrected): TotalDebtToIncomeRatio 0.81, MonthlyIncome 0.77, InterestRate 0.42, LoanAmount 0.35. Among categoricals, EducationLevel is strongest (Cramer's V 0.23). MaritalStatus and LoanPurpose are not related to approval.

3. Pipeline architecture

Everything that learns from data sits inside one scikit-learn `Pipeline`, so training folds are the only data any step ever fits on.

1. **Custom transformer `LoanFeatureEngineer`** (stateless). Creates AgeDeviation (|Age - 40|), LoanToIncome, IncomeMismatchFlag, IncomeAtFloor and AssetToLiability, then drops AnnualIncome.

2. **ColumnTransformer** with six branches: skewed numerics (impute with indicator, scale), other numerics (impute, scale), binary pass-through, nominal (impute "Missing", one-hot), binary categorical (one column), and education.
3. **FeatureUnion** inside the education branch: the ordinal code (education order High School < Associate < Bachelor < Master < Doctorate, mode-imputed) plus a MissingIndicator.
4. **Optional feature selection** step (tuned): L1-penalised logistic selector.
5. **Model** (final: Logistic Regression).

The preprocessing produces 52 model-ready columns from 33 raw ones.

3.1 Preprocessing experiments

Each choice was tested with 5-fold out-of-fold predictions on the training set.

Experiment	Result	Decision
Log-transform skewed money features	AUC 0.9897 (on) vs 0.9961 (off); best cost $612 vs $450	**Off.** The decision rule is linear in raw dollars.
Drop AnnualIncome	No change (it duplicates MonthlyIncome)	Drop it.
Add AgeDeviation	Best cost falls from about $522 to $440	**Keep.** The only engineered feature with a real effect.
Income flags, ratio features	No gain	Kept as cheap data-quality safeguards, pruned by selection if useless.
Education missing flag	Best cost $450 (with) vs $461 (without)	Keep.
Imputation of SavingsAccountBalance: median, mean, KNN, iterative	AUC 0.99607 to 0.99608 for all	Median (simplest).

4. Modeling

4.1 Validation strategy

Stratified 80/20 split (16,000 training, 4,000 test, random_state 42). The test set is used once. All comparisons and searches use the same 5-fold stratified cross-validation on the training set.

4.2 Metrics

Metric	Role
Business cost per applicant (custom: $50,000 x false positives + $8,000 x false negatives, divided by applicants)	Final business verdict. Implemented as a scikit-learn scorer.
ROC-AUC	Model selection and tuning (threshold-free ranking quality)
PR-AUC	Honest under class imbalance
Recall and F1	Classification quality at a fixed cut-off

The break-even probability is 50,000 / (50,000 + 8,000) = 0.862: approve only when the model's probability exceeds it. Baselines per applicant: deny everyone $1,912, approve everyone $38,050, coin flip $19,981.

4.3 Model comparison (5-fold cross-validation, cost at the 0.862 threshold)

Model	Validation AUC	Cost per applicant	Train-validation gap
Logistic Regression	**0.9961**	**$456**	0.0003
Voting ensemble (LR, HGB, RF)	0.9949	$695	0.004
Hist Gradient Boosting	0.9906	$712	0.009
Gradient Boosting	0.9862	$899	0.007
Random Forest	0.9766	$1,314	0.023

Model	Validation AUC	Cost per applicant	Train-validation gap
Decision Tree (depth 5)	0.9426	$1,612	0.012
KNN (k=25)	0.9528	$1,719	0.013
Dummy baseline	0.5000	$1,912	0
Decision Tree (unpruned)	0.8431	$3,268	0.157

The data is dominated by roughly linear effects, so the linear model has almost no variance, while tree ensembles must approximate straight boundaries with steps. The unpruned tree is worse than lending to nobody.

4.4 Classification versus regression

RiskScore was used as a regression target (Ridge and Hist Gradient Boosting Regressor) and the predicted score converted to a decision. Results: AUC 0.889 and 0.929, best cost $1,544 and $1,384 per applicant, against 0.996 and $450 for classification. Classification was chosen.

4.5 Hyperparameter tuning (multi-stage)

Stage	Method	Searched	Outcome
1	RandomizedSearchCV, 40 candidates x 5 folds on the Logistic Regression pipeline	C (log scale), L1 or L2, class weighting, log transform on or off, imputation strategy, feature selection (none, L1 C=0.05, L1 C=0.5)	Best CV AUC 0.99625. Log off confirmed. Class weighting did not change AUC but raised cost at 0.862 to about $640 from about $460.
2	GridSearchCV around the stage-1 winner, 5 folds, one-standard-error refit rule	C from 0.1 to about 300, class weight none or balanced	AUC plateaus above C of about 3. Chosen: **C = 1**, AUC 0.99613 versus 0.99626 at the best C. Train-validation gap 0.00026.

Stage	Method	Searched	Outcome
1b	RandomizedSearchCV, 10 candidates x 5 folds on Hist Gradient Boosting	learning rate, iterations, leaf nodes, leaf size, L2	CV AUC 0.99216, cost $636, visible overfitting (train AUC 0.9995). Logistic Regression wins.
3	Threshold search on out-of-fold predictions	thresholds 0.05 to 0.99	Chosen **0.81**, cost $443 per applicant (theory 0.862 gives $459, default 0.50 gives $850).

Final configuration: Logistic Regression, L2 penalty, C = 1, no class weights, no log transform, L1-based feature selector (C = 0.05) that keeps 22 of 52 columns, decision threshold 0.81. The one-standard-error rule picks the simplest model statistically tied with the best, which shrinks coefficients and protects against collinearity.

5. Evaluation on the held-out test set (4,000 applications)

Metric	Value
ROC-AUC	0.9963 (95% bootstrap interval 0.9952 to 0.9972)
PR-AUC	0.9888 (0.9855 to 0.9916)
Precision, recall, F1 at threshold 0.81	0.984, 0.842, 0.908
False approvals, missed good loans	13 and 151
Cost per applicant	about $464 (interval $378 to $560)

5.1 Dollar impact, scaled to 20,000 applications

Strategy	Approved	Net profit
Deny everyone	0%	$0
Approve everyone	100%	about -$722.8M

Strategy	Approved	Net profit
Model at default 0.50	23.8%	about $21.6M
Model at 0.81 (chosen)	**20.4%**	**about $28.9M**
Perfect foresight (ceiling)	23.9%	about $38.2M

The model captures 75.7% of the profit available under perfect foresight. Choosing 0.81 instead of 0.50 is worth about $7.4M per 20,000 applications. Net profit counts +$8,000 per good loan approved and -$50,000 per bad loan approved, which is equivalent to minimising the error cost.

5.2 Segments

Segment AUC is 0.989 to 0.998 in every group, so no group is badly mis-ranked. The model approves slightly less than the historical rate in nearly every segment because of the conservative threshold. Recall is lowest for unemployed applicants and the middle income quartiles (about 0.74) and highest for high earners and applicants over 60 (about 0.89). Costs are highest for Doctorate holders, unemployed applicants and the top income quartile. Small segments (Doctorate 156, widowed 164, over 60 198) are noisy.

5.3 Feature importance

Permutation importance on the test set (drop in AUC): TotalDebtToIncomeRatio 0.115, MonthlyIncome 0.108, InterestRate 0.088, NetWorth 0.029, CreditScore 0.019, then LoanAmount, LengthOfCreditHistory, EducationLevel, BankruptcyHistory and PreviousLoanDefaults (0.004 to 0.010). In the coefficients, CreditScore has a negative sign because InterestRate is computed from it and carries the credit signal, so the pair must be explained together.

6. Fairness and compliance

Attribute	Lowest-approval group	Disparate-impact ratio (model)	Same ratio in historical labels
Age band	18 to 29	0.34	0.37
Education	High School	0.32	0.35

Attribute	Lowest-approval group	Disparate-impact ratio (model)	Same ratio in historical labels
Employment	Unemployed	0.59	0.66
Marital status	Divorced	0.70	0.78

All ratios fall below the four-fifths guideline of 0.80, but the gaps already exist in the historical data and the model amplifies them slightly. Ablation results (cost per applicant): full model $443, without MaritalStatus $443 (no change, the selector had already dropped it), without Age and AgeDeviation $525 (+$82, about 19%). Experience correlates 0.98 with Age, so it remains a partial age proxy.

Recommendations: exclude MaritalStatus permanently. Treat age as a quantified business versus legal decision for Compliance. Review outcome gaps with a fairness team.

7. Limitations

- The label is the **historical decision**, not realised repayment. A "false positive" means approved but the historical rule said deny. Selection bias applies.
- The data is synthetic and generated from a near-linear rule plus noise, which explains the very high AUC and why a linear model wins. Expect lower performance on real data.
- Remaining errors lie near the decision boundary (probabilities 0.5 to 0.95) and look like label noise.
- The $8,000 and $50,000 costs are averages applied to every loan regardless of size.
- InterestRate may be set after approval in production. Confirm it is available at decision time, otherwise remove it and retrain.
- Education missing always means denial in the data. The model copies this, which is a data-capture rule to review.

8. Deployment and monitoring guide

1. **Pilot as decision support.** Auto-approve above about 0.81, send roughly 0.5 to 0.81 to a loan officer with the top five drivers, auto-deny very low scores with adverse-action reasons.

2. **Keep costs in configuration.** The optimal threshold moves with the default loss (about 0.77 at $30k, 0.86 at $80k). Re-tune quarterly.
3. **Fix data capture:** require education, validate AnnualIncome against MonthlyIncome, remove the $15,000 floor and $300,000 cap.
4. **Monitor:** score drift (population stability index), approval rate, realised default rate by score band, and the segment and fairness tables.
5. **Govern:** version the whole pipeline with its threshold as one object, log every score, and run the tuned gradient boosting model as a shadow challenger.

9. Potential improvements

- Obtain repayment outcomes and model true default probability, with loan-size-dependent costs.
- Calibrate probabilities (isotonic or Platt) and add conformal intervals.
- Test a boosted model with monotonic constraints as a champion/challenger.
- Apply reject inference, fairness-constrained training, and SHAP-based reason codes.
- Use nested cross-validation, Bayesian optimisation and time-based validation once application dates exist.
- Add schema, range and income-consistency validation at data ingestion.

10. Reproducing the work

Requirements. Python 3.8 or later with pandas, NumPy, scikit-learn, SciPy, Matplotlib and Seaborn.

Steps. Place `financial_loan_data.csv` next to the notebook (or change `DATA_PATH` in the setup cell, for example `/content/...` in Colab). Run all cells in order. All randomness uses `random_state=42`. The full run takes roughly 10 to 13 minutes on one CPU core.

Windows note. A joblib warning about counting physical cores is harmless. To silence it, add `import os; os.environ["LOKY_MAX_CPU_COUNT"] = "4"` as the first line of the setup cell.

10.1 Notebook map

Part	Notebook steps	Content
1. Business understanding	1.1 to 2.5	Process, stakeholders, error costs, classification versus regression, metrics, custom cost scorer, baseline targets
2. Data understanding	3.1 to 4.4	Data characteristics, distributions, relationships, statistical tests, data quality, feature types, missing patterns, engineering ideas
3. Pipeline development	5.1 to 5.7	Train/test split, custom transformer, ColumnTransformer and Pipeline, FeatureUnion, imputation, encoding, scaling, decision log
4. Iterative modeling	6.1 to 7.5	Nine models, validation strategy, regression comparison, parameter spaces, three-stage tuning, preprocessing tuning, trade-offs, threshold
5. Evaluation and conclusion	8.1 to 10.4	Test metrics, dollar impact, segments, fairness, error analysis, plots, importance, executive summary, recommendations, improvements

Each code cell is followed by a block explaining what it does, why, how to read the output, and the decision taken.

11. Glossary

- **False positive / false negative:** approving a loan that should be denied / denying a creditworthy applicant.
- **ROC-AUC:** probability the model ranks a random approved applicant above a random denied one.
- **PR-AUC:** area under the precision-recall curve, informative when one class is rare.
- **Out-of-fold (OOF) prediction:** a prediction made by a model that never saw that row in training.

- **Target leakage:** a feature that contains information about the label that would not exist at decision time.
- **One-standard-error rule:** choose the simplest model whose score is within one standard error of the best.
- **Disparate-impact ratio:** lowest group approval rate divided by highest; below 0.80 is a common red flag.
- **Ablation:** removing one component to measure its contribution.
- **VIF:** variance inflation factor, a measure of collinearity (above 10 is severe).